To pay more attention to gain better result

ASSEMLY LANGUAGE 01

Objectives

After studying this part, you should be able to:

- Familiarize yourself with the assembly language, a low level language
- Understand how a program is compiled
- Develop some basic console applications

Contents

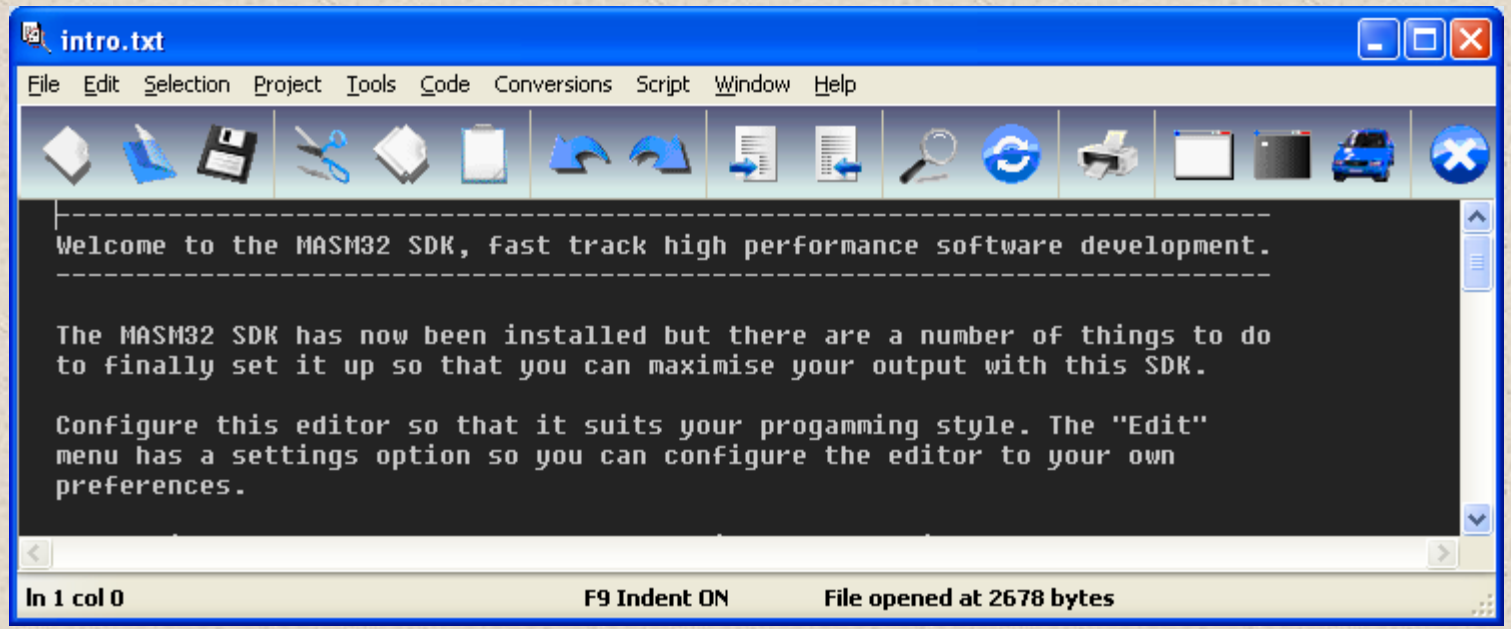
- 1- Install 32/64-bit MASM – MS Macro Assembly
- 2- MASM Integrated Development Environment(IDE)
- 3- Introduction to Microsoft Macro Assembly Language
- Some sample programs

1- Install 32-bit MSAM

- Microsoft Macro Assembler: an x86 assembler that uses the Intel syntax for MS-DOS and Microsoft Windows. Beginning with MASM 8.0 there are two versions of the assembler - one for 16-bit and 32-bit assembly sources, and another (**ML64**) for 64-bit sources only (Wiki)

- Unzip: masm32v11r.zip → Install.exe → Run for installation

- Interface after installation:



Install 32-bit MSAM…

- Installed Contents:

masm32

File Edit View Favorites Tools Help

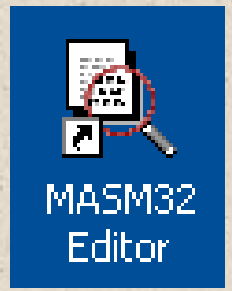
Back Search Folders

Address K:\masm32 Go

Folders

masm32
- bin
- datetime
- examples
- fpulib
- help
- include
- lib
- licence
- m32lib

bin	procs	blockp.ini	jtmake.exe	multitool.exe	se.dll
datetime	script	cg.exe	libbat.qsc	procmap.exe	shellex.exe
examples	testinst	cg.ini	makecimp.exe	prostart.exe	subclass.exe
fpulib	text	cpicker.exe	makecon.qsc	prostart.ini	testbed.qsc
help	tools	dlgmake.exe	makelibs.bat	prostart.set	tmp.qsc
include	tutorial	dlgproc.qsc	makerc.qsc	pths.ini	topgun.chm
lib	vkdebug	dlgtmplt.qsc	maketbl.exe	qeditor.exe	topgun.exe
licence	bigtstbd.qsc	fda32.exe	mangle.exe	qeint.bin	tproc.exe
m32lib	bin2dbex.exe	fda.exe	menuedit.dll	qetb.exe	tproc.txt
macros	blankdlg.qsc	getcolor.exe	menus.ini	qetxt.bin	tview.exe
plugins	bldmakit.qsc	intro.txt	mnutoasm.exe	qsc.dll	ueint.bin

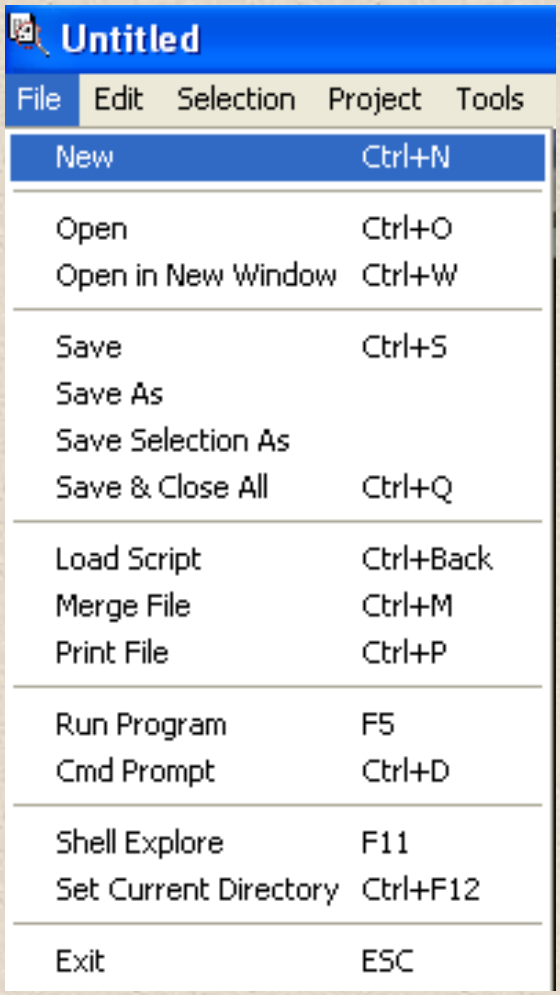


Desktop Icon of MASM, executable file: **qeditor.exe**

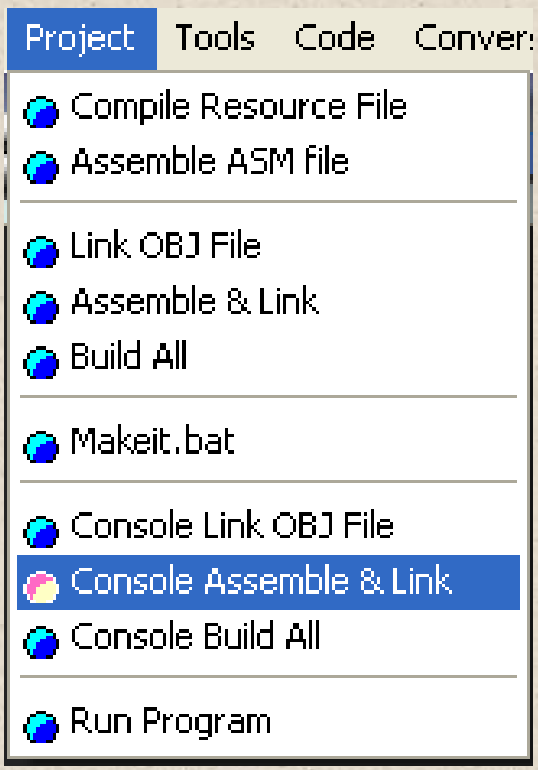
Compiler: bin/ml.exe (32 bit), ml64.exe (64 bit)

You should create a folder as a storage of your exercises

2- MASM Integrated Development Environment

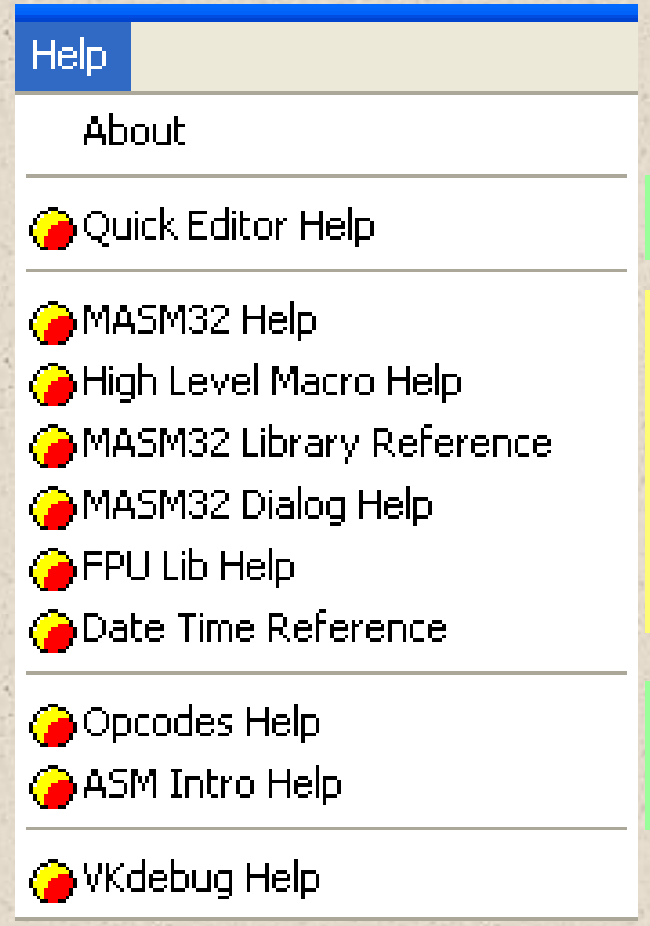


Menu **file** allows user working with files, run program, …



Menu **Project** allows user compiling program, …

2- MASM Integrated Development Environment



Menu **Help** allows user referencing to relative topics:

Using editor

Build-in Libraries in MASM

Opcodes of Intel CPU
Syntaxes of MASM language

3- Introduction to MS Assembly

Demo 1: Write an Assembly program that displays the string 'Hello World' on the screen:

The **flat memory model** is a *non-segmented* configuration available in 32-bit operating systems.

```
; «« Comment begins with ';' to the end of a line
; From masm32\tutorial\console\demo1
;
; Build this with the "Project" menu using
; "Console Assemble and Link"
; «««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

  .486                                        ; create 32 bit code
  .model flat, stdcall                        ; 32 bit memory model
  option casemap :none                        ; case sensitive

  include \masm32\include\windows.inc         ; always first
  include \masm32\macros\macros.asm           ; MASM support macros

; ------------------------------------------------------------------
; include files that have MASM format prototypes for function calls
; ------------------------------------------------------------------
  include \masm32\include\masm32.inc
  include \masm32\include\gdi32.inc
  include \masm32\include\user32.inc
  include \masm32\include\kernel32.inc
; ------------------------------------------------
; Library files that have definitions for function exports
; and tested reliable prebuilt code.
; ------------------------------------------------
  includelib \masm32\lib\masm32.lib
  includelib \masm32\lib\gdi32.lib
  includelib \masm32\lib\user32.lib
  includelib \masm32\lib\kernel32.lib

  .code                          ; Tell MASM where the code starts

  start:                         ; The CODE entry point to the program
      print chr$("Hello world!",13,10) ; 13: carriage return, 10: new line
  exit                           ; exit the program

; ------------------------------
  end start                      ; Tell MASM where the program ends
```

Directives helps the program will conform to Windows

How to create it? NEXT SLIDE

Form of a MASM program

EX01_Hello.asm

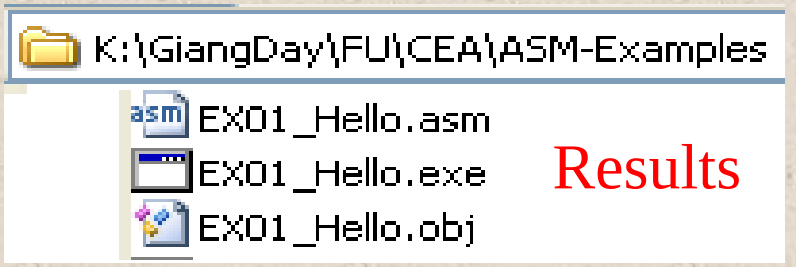
Step 1: Open MASM/ Menu File/ New
Step 2: Copy and paste the code in the next slide to it's editor
Step 3: Save file/EX1_Hello.asm
Step 4: Menu Project/ Console Assemble&Link
Step 5: View results in containing folder
Step 6: Run the program: Click the EX01_Hello.exe
A black window will show then disappear because there is no code to block it.

An Assembly source code is a file whose extension MUST BE .ASM

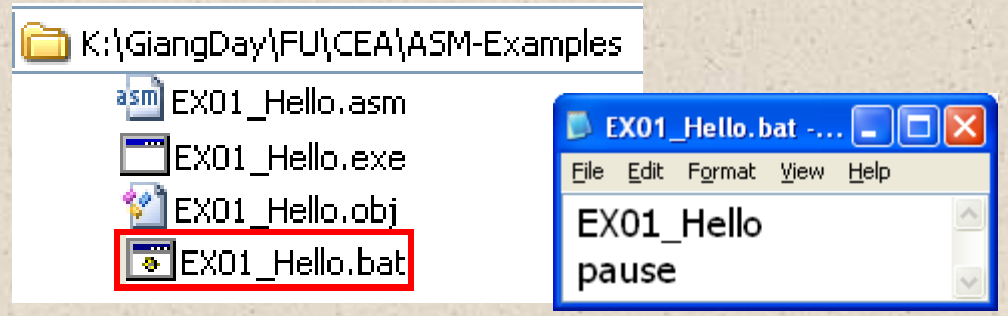
What is the result of compilation?
You can see them in the folder containing you ASM files

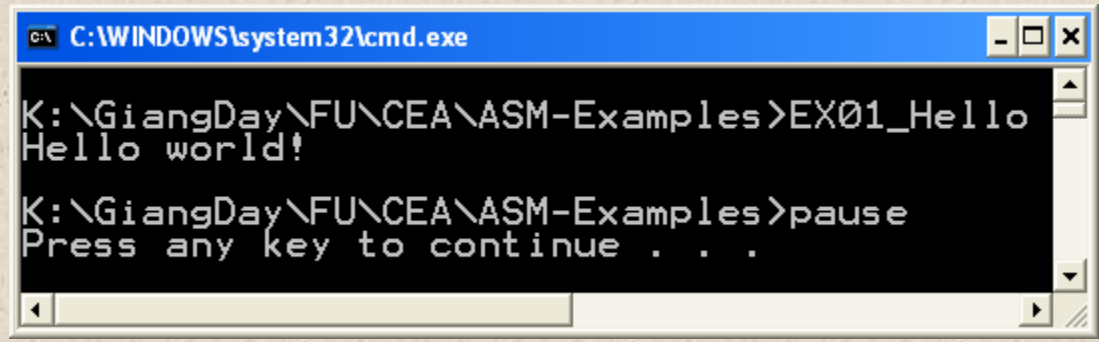
EX01_Hello.asm - Code

```
; ««« Comment begins with ';' to the end of a line
; From masm32\tutorial\console\demo1
;
; Build this with the "Project" menu using
; "Console Assemble and Link"
; «««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

    .486                            ; create 32 bit code
    .model flat, stdcall            ; 32 bit memory model
    option casemap :none            ; case sensitive

    include \masm32\include\windows.inc      ; always first
    include \masm32\macros\macros.asm        ; MASM support macros

  ; -----------------------------------------------------------------
  ; include files that have MASM format prototypes for function calls
  ; -----------------------------------------------------------------
    include \masm32\include\masm32.inc
    include \masm32\include\gdi32.inc
    include \masm32\include\user32.inc
    include \masm32\include\kernel32.inc
  ; ------------------------------------------------
  ; Library files that have definitions for function exports
  ; and tested reliable prebuilt code.
  ; ------------------------------------------------
    includelib \masm32\lib\masm32.lib
    includelib \masm32\lib\gdi32.lib
    includelib \masm32\lib\user32.lib
    includelib \masm32\lib\kernel32.lib

    .code                   ; Tell MASM where the code starts

    start:                  ; The CODE entry point to the program
      print chr$("Hello world!",13,10) ; 13: carriage return, 10: new line
    exit                    ; exit the program

  ; -------------------------------
    end start               ; Tell MASM where the program ends
```

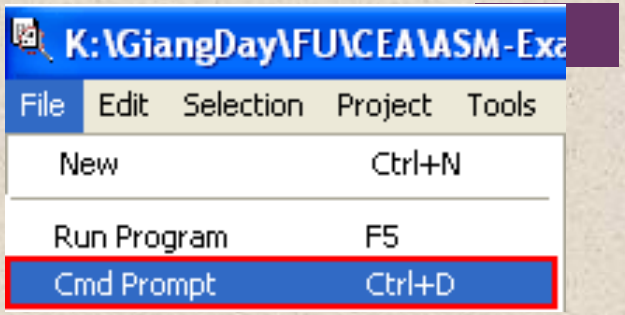


How to run the program and we can see it? ➔ Create EX01_Hello.bat file the run it.





Run a program using the Menu File/Cmd Prompt




```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

K:\GiangDay\FU\CEA\ASM-Examples>dir *.exe
 Volume in drive K is TaiLieu30
 Volume Serial Number is 54D8-4B2E

 Directory of K:\GiangDay\FU\CEA\ASM-Examples

04/16/2015  10:46 AM             2,560 EX01_Hello.exe
04/16/2015  11:41 AM             2,560 EX02_ProcDemo.exe
04/16/2015  01:06 PM             2,560 EX03_Data.exe
04/16/2015  01:46 PM             2,560 EX04_Locals.exe
04/16/2015  03:21 PM             3,072 EX05_Numbers.exe
04/16/2015  04:39 PM             3,072 EX06_Sum.exe
04/16/2015  08:17 PM             3,072 EX07-Swap2.exe
04/16/2015  06:16 PM             3,072 EX07_swap1.exe
04/17/2015  07:17 AM             2,560 MuaXuanChin.exe
               9 File(s)        25,088 bytes
               0 Dir(s)   7,383,755,776 bytes free

K:\GiangDay\FU\CEA\ASM-Examples>EX01_hello
Hello world!

K:\GiangDay\FU\CEA\ASM-Examples>_
```

The command **dir *.exe** will show all exe files stored in the current folder.

Run an application by it's file name (.exe can be ignored)

EX02_ProcDemo.asm

Procedures are a fundamental building block of programs that are build directly into the processor using CALL and RET instructions. This shows how simple it is to do in MASM.

This code is in the directory
masm32\tutorial\console\demo2\Proc.asm

Procedure syntax

```
    .code                           ; Tell MASM where the code starts

; <<<
```

EX02_ProcDemo.asm- Source/Run

```
; From masm32\tutorial\console\demo2
;    Build this with the "Project" menu using
;       "Console Assemble and Link"
; ««««««««««««««««««««««««««««««««««««««««««««
   .486                          ; create 32 bit code
   .model flat, stdcall                  ; 32 bit memory model
   option casemap :none                  ; case sensitive

   include \masm32\include\windows.inc     ; always first
   include \masm32\macros\macros.asm
  ; ---------------------------------------------------------------
  ; include files for function calls
  ; ---------------------------------------------------------------
   include \masm32\include\masm32.inc
   include \masm32\include\gdi32.inc
   include \masm32\include\user32.inc
   include \masm32\include\kernel32.inc
  ; ----------------------------------------------
  ; Library files that have definitions for function
  ; exports and tested reliable prebuilt code.
  ; ----------------------------------------------
   includelib \masm32\lib\masm32.lib
   includelib \masm32\lib\gdi32.lib
   includelib \masm32\lib\user32.lib
   includelib \masm32\lib\kernel32.lib
```

```
 .code                    ; Tell MASM where the code starts

; ««««««««««««««««««««««««««««««««««

start:              ; The CODE entry point to the program
   call main                ; branch to the "main" procedure
   exit
; ««««««««««««««««««««««««««««««««««

main proc
   print chr$("Hi, I am in the 'main' procedure",13,10)
   ret                      ; return to the next instruction after
            "call"
main endp

; ««««««««««««««««««««««««««««««««

end start                   ; Tell MASM where the program
            ends
```

```
K:\GiangDay\FU\CEA\ASM-Examples>EX02_ProcDemo
Hi, I am in the 'main' procedure

K:\GiangDay\FU\CEA\ASM-Examples>_
```

Comments in MASM

Comments are ignored by the assembler

```
; From K:\masm32\tutorial\console\demo3
;                 Build this with the "Project" menu using
;                       "Console Assemble and Link"

comment * <<<
```

EX03_Data.asm

Print a string declared in the program using the operator OFFSET.
The OFFSET operator tells MASM that the text data is at an OFFSET within the file which means in this instance that it is in the **.DATA** section.

```
    .data
      txtmsg db "I am data in the initialised data section",0

    .code                          ; Tell MASM where the code starts
; <<<
```

Basic Data Types in MASM32

Type	Abbr	Size (bytes)	Integer range	Types Allowed
BYTE	**DB**	1	-128.. 127	Character, string
WORD	**DW**	2	-32768..32767	16-bit near ptr, 2 characters, double-byte character
DWORD	**DD**	4	-2Gig..(4Gig-1)	16-bit far per, 32-bit near ptr, 32-bit long word
FWORD	**DF**	6	--	32-bit far ptr
QWORD	**DQ**	8	--	64-bit long word
TBYTE	**DT**	10	--	BCD, 10-byte binary number
REAL4	**DD**	4	--	Single-precision floating number
REAL8	**DQ**	8	--	Double-precision floating number
REAL10	**DT**	10	--	10-byte floating point

D: Defined

Initialized data has this form:

```
.data
  var1  dd  0          ; 32 bit value initialized to zero
  var2  dd  125        ; 32 bit value initialized to 125
  txt1  db  "This is text in MASM",0  ; Initialize a  NULL string
  array dd 1,2,3,4,5,6,7,8   ; array of 8 initialized elements
```

Uninitialized data has this form:

```
.data?
  udat1 dd ?      ; Uninitialized single 32 bit space
  buffa db 128 dup (?)          ; buffer 128 bytes
```

EX03_Data.asm - Source

```
; From K:\masm32\tutorial\console\demo3
;               Build this with the "Project" menu using
;                   "Console Assemble and Link"

    .486                            ; create 32 bit code
    .model flat, stdcall                ; 32 bit memory model
    option casemap :none                ; case sensitive

    include \masm32\include\windows.inc     ; always first
    include \masm32\macros\macros.asm       ; MASM support macros
    include \masm32\include\masm32.inc
    include \masm32\include\gdi32.inc
    include \masm32\include\user32.inc
    include \masm32\include\kernel32.inc
     includelib \masm32\lib\masm32.lib
    includelib \masm32\lib\gdi32.lib
    includelib \masm32\lib\user32.lib
    includelib \masm32\lib\kernel32.lib

    .data
      txtmsg db "I am data in the initialised data section",0

    .code                       ; Tell MASM where the code starts
; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««
start:                          ; The CODE entry point to the program
    call main                   ; branch to the "main" procedure
    exit
; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««
main proc
    print OFFSET txtmsg
    ret                         ; return to the next instruction after "call“
main endp
; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

end start                       ; Tell MASM where the program ends
```

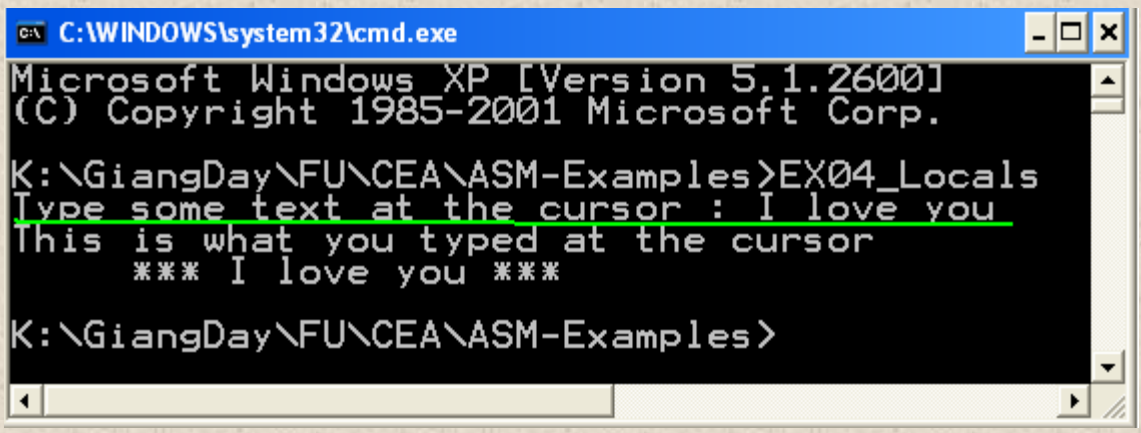
EX04_Locals.asm

How to use of LOCAL variables declared in a procedure?
When the procedure is called, these variables are allocated in program's stack

DECLARE LOCAL VARIABLES

```
LOCAL MyVar:DWORD        ; allocate a 32 bit space on the stack
LOCAL Buffer[128]:BYTE   ; allocate 128 BYTEs of space for TEXT data.
```

How to PROTOTYPE and implement a procedure along with it's parameters?
How to call user-defined procedure?
How can program receive input from user? → Build-in function: input("warning:")



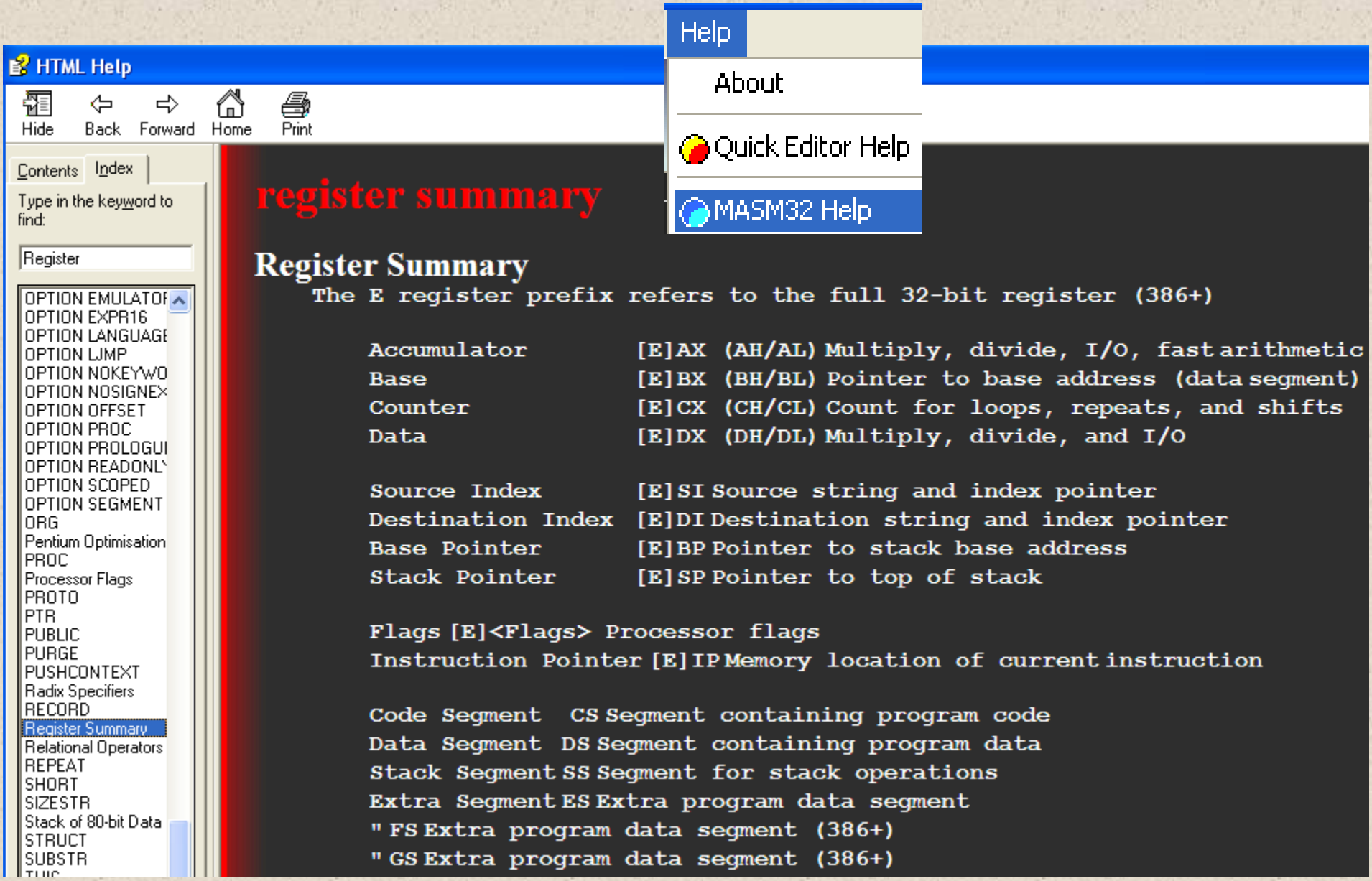
EX04_Locals.asm

(Source code)

```
; Source code From masm32\tutorial\console\demo4\locals.asm
    .486                        ; create 32 bit code
    .model flat, stdcall        ; 32 bit memory model
    option casemap :none        ; case sensitive

    include \masm32\include\windows.inc     ; always first
    ; MASM support macros

    show_text PROTO :DWORD      ;  prototype a method + type of parameter

    .code                       ; Tell MASM where the code starts

; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

start:                          ; The CODE entry point to the program
    call main                   ; branch to the "main" procedure
    exit

; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

main proc
    LOCAL txtinput:DWORD        ; a "handle" for the text returned by "input"
    mov txtinput, input("Type some text at the cursor : ") ; get input string
    invoke show_text, txtinput  ; show inputted string
    ret
main endp

; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

show_text proc string:DWORD
    print chr$("This is what you typed at the cursor",13,10,"     *** ")
    print string                ; show the string at the console
    print chr$(" ***",13,10)
    ret
show_text endp

; ««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

end start                       ; Tell MASM where the program ends
```


```
ype a method + type of parameter

ne code starts

t to the program
" procedure

«««««««««««««««««««««««««««««««««««««««««««««««««««

e" for the text returned by "input"
e cursor : ") ; get input string
d string

«««««««««««««««««««««««««««««««««««««««««««««««««««

cursor",13,10,"     *** ")
ne console
```

Intel CPU 32-bit Registers



Intel CPU Registers

64-bit register	Lower 32 bits	Lower 16 bits	Lower 8 bits
rax	eax	ax	al
rbx	ebx	bx	bl
rcx	ecx	cx	cl
rdx	edx	dx	dl
rsi	esi	si	sil
rdi	edi	di	dil
rbp	ebp	bp	bpl
rsp	esp	sp	spl
r8	r8d	r8w	r8b
r9	r9d	r9w	r9b
r10	r10d	r10w	r10b
r11	r11d	r11w	r11b
r12	r12d	r12w	r12b
r13	r13d	r13w	r13b
r14	r14d	r14w	r14b
r15	r15d	r15w	r15b

CS Code Segment
DS: Data Segment
SS: Stack Segment

https://msdn.microsoft.com/en-us/library/windows/hardware/ff561499(v=vs.85).aspx

EX05-Numbers.asm

(1) How to receive numbers from user?

Raw data from keyboard are string. The function **sval(string)** will convert num-string to signed number.

(2) How to perform a simple addition using registers

add reg1, reg2 will accumulate value in reg2 to reg1

(3) How to print value in a register/variable to screen

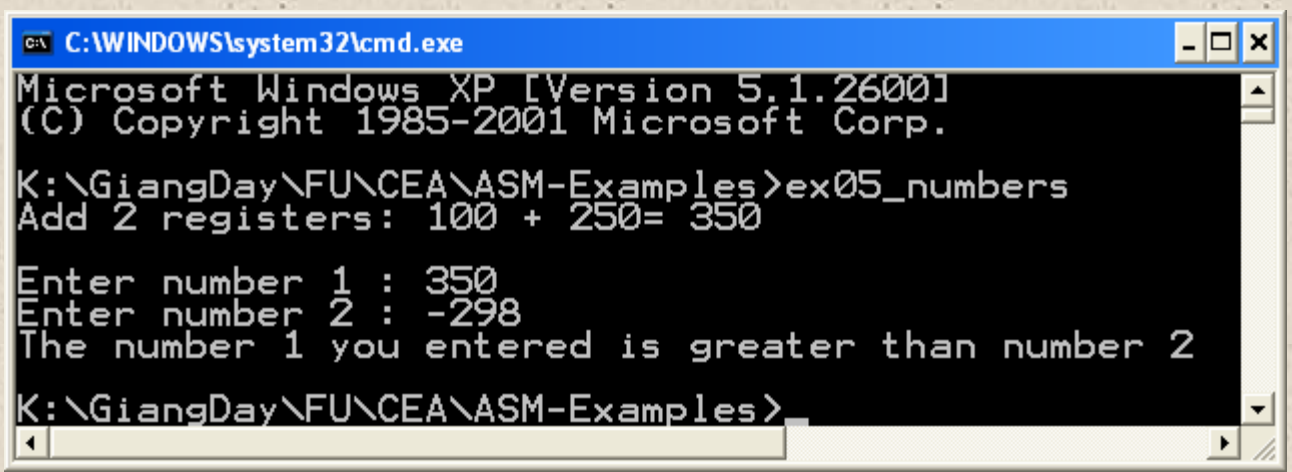
Function **str$(number)** → num-string

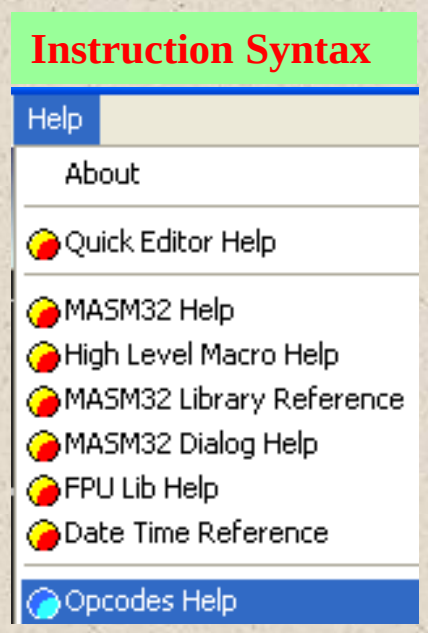
(4) How to compare a memory variable to an immediate number

Use the instruction CMP reg, reg/ CMP reg, var/ CMP var, reg/ CMP mem, immed/ CMP reg, immed (immed= immediate value)

(5) How to branching to different labels after camparation?

Use **j**umps: **JE** (equal), JG (greater than), JL (less than)





EX05-Numbrers.asm

Variables Declarations, Input data, Converting data types

```
; EX05_Numbers.asm

; Declare program model and all libraries using only one file

  include \masm32\include\masm32rt.inc

.code
start:                          ; The CODE entry point to the program
    call main                   ; branch to the "main" procedure
    exit
; <<<
```

EX05-Numbers.asm

Comparing and Branching

```
  ; compare 2 variables and process the result
    mov eax, var1                ; copy var1 to eax
    cmp eax, var2                ; CMP REG, VAR
    je equal                     ; jump if var1 is equal to 100 to "equal"
    jg bigger                    ; jump if var1 is greater than 100 to "bigger"
    jl smaller                   ; jump if var1 is less than 100 to "smaller"

  equal:
    print chr$("2 numbers you entered are equal.",13,10)
    jmp over

  bigger:
    print chr$("The number 1 you entered is greater than number 2",13,10)
    jmp over

  smaller:
    print chr$("The number 1 you entered is smaller than number 2",13,10)

  over:
    ret
main endp
; <<<
```

EX05-Numbers.asm

Source code

```
; EX05_Numbers.asm
; Declare program model and all libraries using only one file

  include \masm32\include\masm32rt.inc

.code
start:                    ; The CODE entry point to the program
    call main             ; branch to the "main" procedure
    exit
; «««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««
main proc
    LOCAL var1:DWORD      ; 2 DWORD integral variables
    LOCAL var2:DWORD      ;
    LOCAL str1:DWORD      ; a string handle for the input data

  ; test the MOV and ADD instructions
    print chr$("Add 2 registers: 100 + 250= ")
    mov eax, 100          ; copy the IMMEDIATE number 100 into the EAX register
    mov ecx, 250          ; copy the IMMEDIATE number 250 into the ECX register
    add ecx, eax          ; ADD EAX to ECX
    print str$(ecx)       ; show the result at the console
    print chr$(13,10,13,10)   ; 2 empty lines

  ; Input 2 integers
    mov var1, sval(input("Enter number 1 : "))
    mov var2, sval(input("Enter number 2 : "))
```

```
  ; compare 2 variables and process the result
    mov eax, var1         ; copy var1 to eax
    cmp eax, var2         ; CMP REG, VAR
    je equal              ; jump if var1 is equal to 100 to "equal"
    jg bigger             ; jump if var1 is greater than 100 to "bigger"
    jl smaller            ; jump if var1 is less than 100 to "smaller"

  equal:
    print chr$("2 numbers you entered are equal.",13,10)
    jmp over

  bigger:
    print chr$("The number 1 you entered is greater than number 2",13,10)
    jmp over

  smaller:
    print chr$("The number 1 you entered is smaller than number 2",13,10)

  over:
    ret
main endp
; «««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««««

end start                 ; Tell MASM where the program ends
```

Exercises

Ex1: Use the **Opcodes help** of the menu Help, describe syntaxes of following MASM instructions

(1) ADD

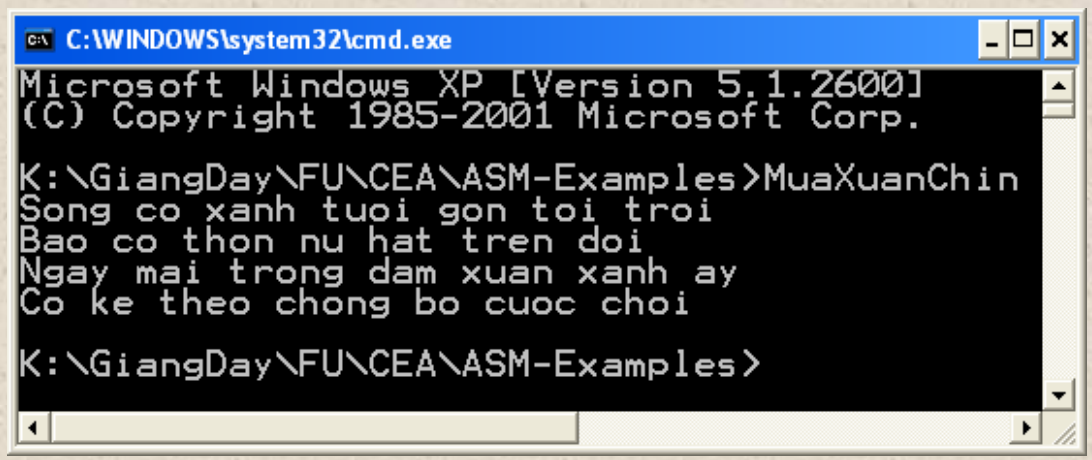
(2) SUB

(3) MUL, IMUL

(4) DIV, IDIV. Which register will store the remainder ?

Ex2:

Write a MASM program that will print the following cantor of Hàn Mặc Tử



Exercises

Ex3: Write a program that will accept 3 numbers, then sum of them and their average will be printed out.

Ex4: Write a MASM program that will solve the equation ax+b=0

Ex5: Write a MASM program that will accept 2 numbers, v1, v2 then print out v1+v2, v1-v2, v1*v2, v1/v2.
Attention: The case in which v2=0 must be managed.

Summary

- Form of a MASM program
- Variable declarations: DB, DD, DW, …
- Basic input, output operations: print, chr$(…), str$(…)
- Data type conversion: sval(..),
- Procedure with parameters: CALL, INVOKE
- Instructions: MOV, ADD, CMP, JE, JG, JL